

Joey Jaehyeok Kim

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Education

Georgia Institute of Technology

M.S. Mechanical Engineering

Aug 2026 – May 2027 (Expected)

B.S. Mechanical Engineering | GPA: 3.78/4.00

Jun 2022 – Jul 2026 (Expected)

Experience

Tesla

Lathrop, California

Test Engineer

Jan 2025 – Jul 2025

- Diagnosed and troubleshot electronic subassembly failures through test program execution, data analysis, and fault injection
- Achieved 98% FPY and 10x mating cycle by leading end-to-end concept to deployment of pogo-pin harness connectors with full tolerance stack-up and GD&T documentation
- \$20,000 procurement cost reduction by developing the design and maintaining uninterrupted inflow of ESD-safe fixtures for critical electronic components
- 4% increase in end-of-line FPY delivered by identifying production constraints through root cause analysis and by integrating corrective actions across Maintenance and Production teams
- Partnered with Supply Chain to decrease downtime by forecasting demand of critical test components, ensuring on-time deliveries, and implementation to production lines
- Collaborated with Manufacturing to uphold ASME Y14.5 documentation standards for custom hardware, ensuring manufacturability and reliability

Cognosos

Atlanta, Georgia

Mechanical Engineer

Jan 2024 – Aug 2024

- Supported end-to-end RTLS product development spanning hardware design, prototyping, testing, and sourcing
- 5% decrease in in-house prototyping turnaround time by implementing a scheduling system for fabrication requests that eliminated resource conflicts, respected maintenance schedules, and aligned rapid iteration with product release schedules
- 97% pass rate on 300+ units of hand-assembled, soldered, and tested RTLS products with verified 1-year operation
- Drove 6% increase in incoming vendor acceptance rate by improving incoming quality assurance, on-site capability audits, and applying DFM principles to optimize plastic enclosure CAD for efficient and cost-effective injection molding
- Led impact and compression testing, authoring pass/fail criteria for all RTLS mechanical and electrical assemblies
- Managed full SolidWorks library migration to Onshape while overseeing schematic creation ensuring ASME Y14.5 compliance

Computational Design and Metamaterials Lab

Atlanta, Georgia

Research Assistant under Dr. Bolei Deng

Aug 2024 – Present

- Validating TPMS models through shear and axial testing to correlate experimental stress-strain data with simulation outcomes
- Creating a differentiable simulation that can optimize metamaterial structure to replace padding in motorcycle headgear
- Performed material testing and data analysis to correlate simulation predictions with real-world deformation behavior
- Designed a tube with both dilative and compressive properties as an alternative to surgical ETTs
- Modeled bistable unit cells to create metamaterials with 2 DoF on flat surface and neutral buoyancy

Projects

Helical Compression Spring Design Optimization

Mar 2026 – Apr 2026

- Constrained nonlinear optimization to minimize helical compression spring mass, defining 3 design variables and 7 engineering constraints spanning shear stress, deflection, surge avoidance, manufacturability, and buckling limits
- Implemented the solver in MATLAB using fmincon (interior-point algorithm) with multi-start Latin hypercube sampling, achieving a 21.1 g minimum-mass design with stress and spring index constraints simultaneously active at optimality
- Validated global optimality through a 40x40 sweep, confirming convergence to the same design point and generating mass contour, Pareto, and spring rate surface visualizations
- Conducted sensitivity analysis revealing allowable shear stress as the dominant design driver—a 10% relaxation of the Zimmerli stress criterion yielded an 18% mass reduction

CPU/GPU Thermal Lifecycle Simulator

Feb 2026 – Mar 2026

- 2D finite difference thermal simulation of a multi-layer CPU/GPU chip package (silicon, TIM, copper heat spreader) modeling transient heating and cool-down phases to predict peak junction temperatures and identify thermal bottlenecks
- Implemented and cross-validated explicit, implicit, and steady-state Gauss-Seidel solvers—achieving agreement within 0.5K and confirming TIM as the dominant thermal resistance while the copper spreader effectively redistributes heat laterally
- Simulated cool-down dynamics, tracking thermal decay to ambient and revealing asymmetric cooling rates between layers due to differing thermal mass, with peak temperature history characterizing the package's thermal time constant

TPMS for Headgear Impact Protection

Sep 2025 – Present

- Differential simulation based on Cauchy momentum equation and Hooke's law to compute strain-energy gradients to inverse design metamaterial structures with optimized 3D cell geometry
- Achieved 7% greater shear strength and improved strength-to-weight efficiency through FEA-driven comparisons between conventional helmet foam and TPMS lattice liners
- Simulate heat transfer through TPMS geometries to analyze airflow and temperature at transient and steady states
- Maintain $HIC < 250$ and $BrIC < 0.45$ and meet DOT impact requirements through iterative tuning of lattice parameters

Voice-Controlled Robotic Arm Assistant

Jul 2025 – Present

- 6-DOF robotic arm with CAD and integrated AI voice assistant
- Motion control via Python with logging and unit tests for reliability
- DC motors, IR sensors, solenoids, AC adapters, potentiometers, and pneumatics to diversify robot capabilities

Detecting Faulty PCBs using Thermal Image Analysis

May 2025 – Jun 2025

- Thermal defect detection pipeline using Gaussian smoothing, morphological operations, and region-based segmentation to isolate abnormal heat signatures
- 98% classification accuracy by combining image processing with synthetic thermal-image augmentation
- Used diffusion model to generate synthetic thermal image datasets to model rare failure modes

Skills

- **Design:** SOLIDWORKS, CATIA, Onshape, AutoCAD, GD&T (ASME Y14.5), DFM, FMEA, tolerance stack-up
- **Simulation & Analysis:** FEA, Finite-difference methods, thermal modeling, structural mechanics
- **Test & Manufacturing:** NPI, DVT, RCA, Vector CANape, LabView, Siemens HMI, Arena PLM
- **Programming:** Python, MATLAB, C++, Java, Go, Swift, TP Programming
- **Fabrication:** 3D printing, laser cutting, soldering, lathe, bandsaw, drill press